Study notes: Abstract Algebra I, Module 1

Fraleigh, *A First Course in Abstract Algebra* (8th ed.), Module 1 (sections 1 and 2). Numbers are the book's own. Tables only, no figures. Exercises (printed pp. 17–19 and 28–31) are not included. Student notes in my own words; every definition, theorem, example and table is kept with its book number.

Section 1: Binary operations

Fraleigh 8th ed., printed pp. 11–16 (1.1–1.30)

Before you start

- A **set** is a collection of objects. $S \times S$ is the set of all ordered pairs (a, b) with $a, b \in S$ (Definition 0.4 in the book).
- A **function** sends each input to one output. $f \circ g$ means "first g, then f": $(f \circ g)(x) = f(g(x))$.
- Arithmetic laws you already use: $a + b = b + a$, $a + 0 = a$, $a + (-a) = 0$; the same laws for $\times$ with 1 and $\frac{1}{a}$.
- Closure of a subset (Section 1.4) and the warnings in 1.26–1.30 are new. Learn the idea before reading them.

1. Binary operations

1.1 What an operation is

A **binary operation** $*$ on a set S is a rule that takes an ordered pair (a, b) of elements of S and gives exactly one element of S, written $a * b$. The word "binary" means the rule takes *two* inputs. The book calls it an operation when there is no risk of confusion.

Addition $+$ on $\mathbb{R}$ is an operation. Multiplication $\cdot$ on $\mathbb{R}$ is a different operation on the same set. The same two rules work on $\mathbb{C}$, $\mathbb{Z}$, $\mathbb{R}^+$, and $\mathbb{Z}^+$ (Example 1.2).

> **DEFINITION 1.1**
> A binary operation $*$ on S is a function from $S \times S$ into S. For each $(a, b) \in S \times S$, the element $*((a, b))$ is written $a * b$.

Note that the operation must be defined for *every* ordered pair of elements of S. Matrix addition on $M(\mathbb{R})$ is *not* an operation on that set, because $A + B$ is not defined when A and B have different sizes (Example 1.3).

1.2 Closed subsets and induced operations

Sometimes an operation on S gives an operation on a smaller set $H \subseteq S$. The question is whether the answer stays inside H.

> **DEFINITION 1.4**
> Let $*$ be an operation on S and $H \subseteq S$. H is **closed under** $*$ if $a * b \in H$ for all $a, b \in H$. Then the operation restricted to H is the **induced operation**.

The set S is always closed under its own operation. A subset may not be.

Example 1.5. Ordinary addition on $\mathbb{R}$ does not induce an operation on $\mathbb{R}^*$ (the nonzero reals): 2 and -2 are in $\mathbb{R}^*$, but $2 + (-2) = 0 \notin \mathbb{R}^*$.

Example 1.6. Let $H = \{n^2 \mid n \in \mathbb{Z}^+\}$.

- Addition: $1^2 = 1$ and $2^2 = 4$ are in H, but $1 + 4 = 5 \notin H$. So H is not closed under addition.
- Multiplication: if $r = n^2$ and $s = m^2$, then $rs = (nm)^2$, and $nm \in \mathbb{Z}^+$. So H is closed under multiplication.

The method: write what membership in H means for r and s, then check whether the result has that form.

Example 1.7. Let F be the set of real-valued functions with domain $\mathbb{R}$. For $f, g \in F$ define, for each $x \in \mathbb{R}$:

$$(f + g)(x) = f(x) + g(x), \quad (f - g)(x) = f(x) - g(x), \quad (f \cdot g)(x) = f(x)g(x),$$

Each of these is again a real-valued function on $\mathbb{R}$, so F is closed under all four operations.

In this text we abstract these familiar structures. Unfamiliar examples help show which features are essential.

1.3 Three examples of operations

Example 1.8. On $\mathbb{Z}^+$ let $a * b$ be the smaller of a and b (or the common value if $a = b$). For example $2 * 11 = 2$, $15 * 10 = 10$, and $3 * 3 = 3$.

Example 1.9. On $\mathbb{Z}^+$ let $a *' b = a$. For example $2 *' 3 = 2$ and $25 *' 10 = 25$.

Example 1.10. On $\mathbb{Z}^+$ let $a *'' b = (a * b) + 2$, where $*$ is from Example 1.8. For example $4 *'' 7 = 6$, $25 *'' 9 = 11$, and $6 *'' 6 = 8$.

These look artificial, but Example 1.8 is used all the time: a GPS route planner, for example, keeps the shorter of two routes. Notice that Example 1.8 gives the same answer in either order. Example 1.9 does not: $5 *' 7 = 5$, but $7 *' 5 = 7$. That difference leads to the next definition.

1.4 Commutativity

> **DEFINITION 1.11**
> An operation $*$ on S is **commutative** if $a * b = b * a$ for all $a, b \in S$.

The book's convention: definitions are if-and-only-if statements, but theorems are not.

> **PLUS TWO RECALL: COMMUTATIVE LAW**
> **Recall.** Addition and multiplication of numbers are commutative.
> **Re-derive.**
> 1. Take $a = 2$, $b = 3$. Then $2 + 3 = 5$ and $3 + 2 = 5$. Equal.
> 2. Take $a = 2$, $b = 3$ for subtraction: $2 - 3 = -1$ but $3 - 2 = 1$. Not equal.
> 3. So subtraction is not commutative, and commutativity has to be checked for each operation.
>
> **Saved.** Commutativity is a property of an operation, not of a set. $*$ on $\mathbb{Z}^+$ in Example 1.8 is commutative; $*'$ in Example 1.9 is not (Definition 1.11).

1.5 Associativity

A binary operation combines two elements at a time. For three elements a, b, c there are two obvious ways to combine them: $(a * b) * c$ and $a * (b * c)$.

With $*$ from Example 1.8: $(2 * 5) * 9 = 2 * 9 = 2$, and $2 * (5 * 9) = 2 * 5 = 2$. They agree. In general, for this $*$,

$$(a * b) * c = a * (b * c),$$

so the expression $a * b * c$ has one meaning.

For $*''$ from Example 1.10 they do not agree:

$$(2 *'' 5) *'' 9 = 4 *'' 9 = 6, \qquad 2 *'' (5 *'' 9) = 2 *'' 7 = 4.$$

So $a *'' b *'' c$ is ambiguous.

> **DEFINITION 1.12**
> An operation on S is **associative** if $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$.

If $*$ is associative, longer expressions such as $a * b * c * d$ are not ambiguous: the book states (without proof) that brackets may be put in any way, and the result is the same.

> **PLUS TWO RECALL: ASSOCIATIVE LAW**
> **Recall.** The associative law for addition and multiplication of numbers.
> **Re-derive.**
> 1. Take $a = 2$, $b = 3$, $c = 4$. $(2 + 3) + 4 = 5 + 4 = 9$.
> 2. $2 + (3 + 4) = 2 + 7 = 9$. Equal.
> 3. Take the same numbers for subtraction: $(2 - 3) - 4 = -5$, but $2 - (3 - 4) = 3$. Not equal.
>
> **Saved.** Addition and multiplication are associative; subtraction is not (Definition 1.12).

1.6 Composition of functions is associative

For any set S and functions f, g, h mapping S into S, the composition $f \circ g$ is again such a function. The next theorem says composition is associative.

> **THEOREM 1.13 (Associativity of composition).** Let S be a set and let f, g, h map S into S. Then $f \circ (g \circ h) = (f \circ g) \circ h$.

Proof. Two functions are equal if they give the same value at every $x \in S$. For $x \in S$:

$$(f \circ (g \circ h))(x) = f((g \circ h)(x)) = f(g(h(x))),$$

$$((f \circ g) \circ h)(x) = (f \circ g)(h(x)) = f(g(h(x))).$$

Both give $f(g(h(x)))$, so the functions are equal. ◆

Matrix multiplication is associative. The book says this is tedious with summation notation. It follows at once from Theorem 1.13 when matrices are read as linear transformations, since matrix multiplication then corresponds to composition.

1.7 Identity elements

The numbers 0 and 1 play special roles: for any real a, $a+0=a$, and $a\times 1=a$. We call 0 the additive identity and 1 the multiplicative identity.

DEFINITION 1.14

Let $*$ be an operation on S. An element $e\in S$ is an **identity element** for $*$ if $a*e=e*a=a$ for all $a\in S$.

Both conditions are required because the operation need not be commutative. If it is commutative, one of the two conditions is enough.

PLUS TWO RECALL: IDENTITY ELEMENTS

Recall. 0 is the additive identity and 1 the multiplicative identity on the reals.

Re-derive.

1. Take any a. $a+0=0+a=a$. So 0 satisfies Definition 1.14 for $+$.
2. $a\cdot 1=1\cdot a=a$. So 1 satisfies Definition 1.14 for $\cdot$.
3. For subtraction, $a-0=a$ holds, but $0-a=-a$, which is not a in general. So 0 is a right identity for subtraction but not a left identity, and Definition 1.14 needs both sides to be checked.

Saved. 0 and 1 are identities for $+$ and $\cdot$ (Definition 1.14).

THEOREM 1.15 (Uniqueness of identity). A set with a binary operation has at most one identity element.

Proof. Suppose e and e' are both identities. Look at $e*e'$. Since e is an identity, $e*e'=e'$. Since e' is an identity, $e*e'=e$. Therefore $e=e'$. ◆

The book notes that the proof uses only the identity property, not any other property of the operation.

1.8 More examples of identities

Example 1.16. Let F be the set of all real-valued functions on $\mathbb{R}$, as in Example 1.7.

- The function $\iota(x)=x$ is the identity for composition. For any $f\in F$, $f\circ\iota=\iota\circ f=f$.
- The function $m(x)=1$ is the identity for multiplication of functions.
- The function $a(x)=0$ is the identity for addition of functions.
- Subtraction of functions has no identity element.

1.9 Inverses

For addition the inverse of a is $-a$. For multiplication the inverse of a nonzero a is $\frac{1}{a}$.

DEFINITION 1.17

Let $*$ be an operation on S with identity e. For $x\in S$, an **inverse** of x is an element x' with $x*x'=x'*x=e$.

PLUS TWO RECALL: INVERSES

Recall. Additive inverse of a is $-a$; multiplicative inverse of $a\neq 0$ is $\frac{1}{a}$.

Re-derive.

1. Take $a=5$. $5+(-5)=0$, the additive identity. So -5 is an inverse of 5 for $+$.
2. $5\cdot\frac{1}{5}=1$, the multiplicative identity. So $\frac{1}{5}$ is an inverse of 5 for $\cdot$.
3. 0 has no multiplicative inverse, because $0\cdot x=0\neq 1$ for every x.

Saved. Inverses are defined relative to an identity (Definition 1.17).

Example 1.18. On the set F of Example 1.16, the operation is composition and the identity is ι. The usual inverse of a function f is a function f' with $f\circ f'=f'\circ f=\iota$. So f has an inverse exactly when f is one-to-one and onto. The two definitions agree.

2. Operations given by tables

For a finite set S we can define an operation by a table. The elements of S are listed across the top and down the left side in the same order. The entry in row i, column j is $(\text{ith entry on the left})*(\text{jth entry on the top})$.

Example 1.19. Table 1.20 defines $*$ on $S=\{a,b,c\}$. Reading the table gives $a*b=c$ and $b*a=a$, so $*$ is not commutative.

1.20 Table

$*$	a	b	c
a	b	c	b
b	a	c	b
c	c	b	a

A table defines a commutative operation exactly when its entries are symmetric about the diagonal from the top left to the bottom right. Entry (i,j) equals entry (j,i).

Example 1.21. Complete Table 1.22 so that $*$ is commutative on $S=\{a,b,c,d\}$.

Solution. Table 1.22 has $b*a=d$. For commutativity $a*b=d$ too, so d goes in the square symmetric to it across the diagonal. Filling the rest the same way gives Table 1.23.

1.22 Table (as given)

$*$	a	b	c	d
a	b			
b	d	a		
c	a	c	d	
d	a	b	b	c

1.23 Table (completed, commutative)

$*$	a	b	c	d
a	b	d	a	a
b	d	a	c	b
c	a	c	d	b
d	a	b	b	c

Example 1.24. When an operation has an identity, the convention is to list the identity first in the heads. Then the first row and first column repeat the head row and head column, as in Table 1.25.

1.25 Table

$*$	e	a	b	c
e	e	a	b	c
a	a	c	c	a
b	b	a	e	c
c	c	e	a	b

3. Warnings: when a rule is not an operation

To define a binary operation $*$ on a set S we must make sure of two things:

1. exactly one element is assigned to each ordered pair of elements of S;
2. for each ordered pair, the element assigned is again in S.

If Condition 1 fails for some pairs, $*$ is **not everywhere defined**. If the rule can give several answers, $*$ is **not well defined**. If Condition 2 fails, S is **not closed** under $*$.

Example 1.26. Test $a*b=a/b$ on three sets.

- On $\mathbb{Q}$: $2*0=2/0$ is undefined. Condition 1 fails, so this is not an operation on $\mathbb{Q}$.
- On $\mathbb{Q}^*$ (no zero): both conditions hold, since a/b is a nonzero rational whenever $a,b\neq 0$. This is an operation.
- On $\mathbb{Z}^+$: $1*2=\frac{1}{2}\notin\mathbb{Z}^+$. Condition 2 fails.

The examples that follow test attempts to define operations. The symbol $*$ stands for the attempted operation each time.

Example 1.27. On F of Example 1.7 let $f*g=h$, where $h(x)=f(x)/g(x)$. This fails Condition 2: g may be zero somewhere. For $f=\cos$ and $g(x)=x^2$, $h(0)$ is undefined, so $h\notin F$.

Example 1.28. On F let $f*g=h$, where h is a function greater than both f and g. This is not well defined: the phrase "greater than" is never defined for functions, and even with a definition there are many such h.

Example 1.29. Let S be a set of 20 people, no two of the same height. Let $a*b=c$, where c is the tallest person in S. This is an operation: it is defined for every pair and always gives one person in S. It is not interesting, but it is legitimate.

Example 1.30. Let $a*b=c$, where c is the shortest person in S who is taller than both a and b. This is not everywhere defined: if a or b is the tallest person,

no such c exists, so $a * b$ is not determined.

Section 2: Groups

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Before you start

- Section 1 gives the four properties we need: closure, associativity, identity and inverses (Definitions 1.4, 1.12, 1.14, 1.17).
- Matrices: $A + B$ needs equal sizes; AB needs the columns of A to match the rows of B. The identity matrix I_n has 1 on the diagonal and 0 elsewhere. A^{-1} exists when $AA^{-1} = A^{-1}A = I_n$.
- Functions and composition from Section 1 (1.7 and 1.13). The inverse function of a one-to-one, onto function.
- Historical notes in Section 2 are background; they are not needed for the definitions.

1. The definition of a group

2.1 Four ideas, three axioms

In school we solve $5 + x = 2$ and $2x = 3$ by the same steps: add -5 to both sides, use associativity, use the identity and inverse properties. Section 2 asks which properties of an operation make those steps work for any set. The answer is three axioms.

> **DEFINITION 2.1** A **group** $\langle G, * \rangle$ is a set G, closed under a binary operation $*$, such that:

- $\mathcal{G}_1$ (associativity): $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.
- $\mathcal{G}_2$ (identity): there is $e \in G$ with $e * x = x * e = x$ for all $x \in G$.
- $\mathcal{G}_3$ (inverses): for each $a \in G$ there is $a' \in G$ with $a * a' = a' * a = e$.

Our solving steps used exactly these properties. Solving $a * x = b$ in a group works the same way as solving $5 + x = 2$.

2.2 Examples: the reals and the nonzero reals

Example 2.2. $\langle \mathbb{R}, + \rangle$ is a group. Its identity is 0, and the inverse of a is $-a$. The set $\langle \mathbb{R}, \cdot \rangle$ is not a group, because 0 has no multiplicative inverse. The set $\langle \mathbb{R}^*, \cdot \rangle$ of nonzero reals is a group: multiplication is associative, 1 is the identity, and every nonzero a has the inverse $\frac{1}{a}$. This is why $2x = 3$ can be solved: the solution $x = \frac{3}{2}$ comes from the inverse $\frac{1}{2}$ in $\mathbb{R}^*$.

> **PLUS TWO RECALL: IDENTITY AND INVERSE FOR $+$ AND $\cdot$**
>
> **Recall.** 0 is the additive identity; 1 is the multiplicative identity; $-a$ and $\frac{1}{a}$ are the inverses.
>
> **Re-derive.**
>
> 1. $a + 0 = a$ and $a + (-a) = 0$, so 0 is the identity and $-a$ the inverse for $+$ on $\mathbb{R}$.
> 2. $a \cdot 1 = a$ and $a \cdot \frac{1}{a} = 1$ for $a \neq 0$, so 1 is the identity and $\frac{1}{a}$ the inverse for $\cdot$ on $\mathbb{R}^*$.
> 3. $0 \cdot x = 0 \neq 1$ for every x, so 0 has no multiplicative inverse, and $\mathbb{R}$ fails Definition 2.1 under $\cdot$.
>
> **Saved.** $\langle \mathbb{R}, + \rangle$ and $\langle \mathbb{R}^*, \cdot \rangle$ are groups (Example 2.2).

It is convenient to say "G is a group under $*$", and to say "$\mathbb{R}$ is a group" when $\mathbb{R}$ is a group under addition and no other operation is meant.

> **DEFINITION 2.3** A group G is **abelian** if its operation is commutative.

Examples 2.4 and 2.5 (not groups).

- $\mathbb{Z}^+$ under addition (2.4): there is no identity for $+$ in $\mathbb{Z}^+$, since $0 \notin \mathbb{Z}^+$.
- The nonnegative integers under addition (2.5): 0 is the identity, but 2 has no inverse, since $2 + x = 0$ needs $x = -2$.

Examples 2.6 to 2.10 (abelian groups).

- 2.6: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ under addition are abelian groups.
- 2.7: $\mathbb{Z}^+$ under multiplication is not a group. 1 is the identity, but 3 has no inverse in $\mathbb{Z}^+$.
- 2.8: $\mathbb{Q}^+$ and $\mathbb{R}^+$ (positive numbers) and $\mathbb{Q}^*$, $\mathbb{R}^*$, $\mathbb{C}^*$ (nonzero numbers) are abelian groups under multiplication.
- 2.9: the real-valued functions with domain $\mathbb{R}$ form an abelian group under function addition.
- 2.10 (linear algebra): the axioms for a vector space V that concern only vector addition say exactly that V is an abelian group under vector addition.

2.3 Examples with matrices

Example 2.11. $M_{m \times n}(\mathbb{R})$, the $m \times n$ real matrices under matrix addition, is an abelian group. The identity is the zero matrix, all of whose entries are 0.

Example 2.12. $M_n(\mathbb{R})$, the $n \times n$ real matrices under matrix multiplication, is *not* a group. The zero matrix has no inverse.

> **PLUS TWO RECALL: MATRICES**
>
> **Recall.** Matrices of the same size add entrywise; AB is defined when the columns of A match the rows of B; I_n is the identity for multiplication of $n \times n$ matrices.
>
> **Re-derive.**
>
> 1. The zero matrix O satisfies $A + O = O + A = A$, so it is the identity for addition (Example 2.11).
> 2. For O in $M_n(\mathbb{R})$: $O \cdot A = O$ for every A. There is no A with $OA = I_n$, so O has no inverse, and $M_n(\mathbb{R})$ is not a group under multiplication (Example 2.12).
> 3. For a 2×2 matrix with $ad - bc \neq 0$, the inverse exists. Section 2.14 uses this condition.
>
> **Saved.** Addition of equal-size matrices gives an abelian group; multiplication of all $n \times n$ matrices does not.

Example 2.13 (a non-abelian group). Let T be the set of isometries of the plane: functions that preserve distance. Examples are translations and rotations. Under composition T is a group:

- Closure: the composition of two distance-preserving maps preserves distance.
- Associativity: Theorem 1.13.
- Identity: the identity map ι.
- Inverse: the inverse function of an isometry is again an isometry.

T is not abelian. Take $\phi(x, y) = (-x, y)$ (reflection across the y-axis) and $\theta(x, y) = (-y, x)$ (rotation by $\frac{\pi}{2}$ about the origin). Then $\phi \circ \theta(1, 0) = \phi(0, 1) = (0, 1)$, but $\theta \circ \phi(1, 0) = \theta(-1, 0) = (0, -1)$. So $\phi \circ \theta \neq \theta \circ \phi$.

Example 2.14. Let S be the set of invertible $n \times n$ real matrices.

- Closure: if A^{-1} and B^{-1} exist, then $(AB)(B^{-1}A^{-1}) = I_n$, so AB is invertible.
- Associativity holds for matrix multiplication; I_n is the identity; each element has an inverse by the definition of S.

So S is a group, and it is not abelian. It is the general linear group of degree n, written $GL(n, \mathbb{R})$. Matrices in $GL(n, \mathbb{R})$ correspond to invertible linear transformations of $\mathbb{R}^n$, and matrix multiplication corresponds to composition, so the same group also appears as $GL(\mathbb{R}^n)$. The two groups have essentially the same operations, and Definition 2.22 makes this precise.

Example 2.15. Let $*$ on $\mathbb{Q}^+$ be $a * b = ab/2$.

- Associativity: $(a * b) * c = \frac{abc}{4}$ and $a * (b * c) = \frac{abc}{4}$.
- Identity: $2 * a = a * 2 = a$, so 2 is the identity.
- Inverse: $a * \frac{4}{a} = \frac{4}{a} * a = 2$, so $a' = \frac{4}{a}$.

So $\langle \mathbb{Q}^+, * \rangle$ is a group. The example is a reminder that the identity need not be 0 or 1.

2. Elementary properties of groups

The next theorems use only Definition 2.1. Each proof uses the earlier results in order.

2.4 Cancellation

> **THEOREM 2.16 (Cancellation laws).** In a group G, $a * b = a * c$ implies $b = c$, and $b * a = c * a$ implies $b = c$, for all $a, b, c \in G$.

Proof. Suppose $a * b = a * c$. Let a' be the inverse of a (axiom $\mathcal{G}_3$). Then $a' * (a * b) = a' * (a * c)$. By associativity, $(a' * a) * b = (a' * a) * c$. By $\mathcal{G}_3$, $a' * a = e$, so $e * b = e * c$. By $\mathcal{G}_2$, $b = c$. The right cancellation law follows in the same way, multiplying on the right by a'. ◆

In real arithmetic, $2a = 2b$ implies $a = b$ by dividing by 2. The proof above is the same argument with the inverse of a in place of $\frac{1}{2}$.

2.5 Linear equations have unique solutions

THEOREM 2.17 If G is a group and $a, b \in G$, then $a * x = b$ and $y * a = b$ have unique solutions x and y in G.

Proof (existence). Let $x = a' * b$. Then

$$\begin{aligned} a * (a' * b) &= (a * a') * b \quad \text{(associativity)} \\ &= e * b \quad \text{(definition of } a') \\ &= b \quad \text{(property of } e). \end{aligned}$$

So $x = a' * b$ is a solution. Similarly $y = b * a'$ solves $y * a = b$.

Proof (uniqueness). Suppose $y_1 * a = b$ and $y_2 * a = b$. Then $y_1 * a = y_2 * a$, and by Theorem 2.16, $y_1 = y_2$. The uniqueness of x follows in the same way. ◆

The solutions $a' * b$ and $b * a'$ need not be equal unless the operation is commutative. The book notes that the proof of uniqueness is the usual method: assume two solutions and show they are equal.

2.6 The identity and inverses are unique

THEOREM 2.18 In a group G, there is only one element e with $e * x = x * e = x$ for all $x \in G$. For each $a \in G$ there is only one a' with $a' * a = a * a' = e$.

Proof. Theorem 1.15 shows the identity is unique; no group axiom beyond closure is needed. For the inverse, suppose a' and a'' are both inverses of a. Then $a * a'' = a * a' = e$. By Theorem 2.16, $a'' = a'$. ◆

COROLLARY 2.19 In a group G, for all $a, b \in G$, $(a * b)' = b' * a'$.

Proof. $(a * b) * (b' * a') = a * (b * b') * a' = a * e * a' = a * a' = e$. By Theorem 2.18 this makes $b' * a'$ the unique inverse of $a * b$. ◆

2.7 Weaker-looking axioms (one-sided)

Some authors use weaker axioms. A **semigroup** is a set with an associative operation; a **monoid** is a semigroup with an identity. Every group is both.

A group can also be defined with one-sided axioms:

1. the operation is associative;
2. there is a left identity e with $e * x = x$ for all x;
3. each a has a left inverse a' with $a' * a = e$.

From these one can prove that a left identity is also a right identity, and a left inverse is also a right inverse. So the one-sided definition gives the same groups. It is not weaker. The book suggests it may sometimes be easier to check the left axioms; by symmetry there are also right axioms.

3. Group isomorphisms and tables

2.8 Groups of one, two and three elements

A one-element set $\{e\}$ with $e * e = e$ is a group. Its identity is its own inverse.

The group $G = \{1, -1\}$ under multiplication has Table 2.20.

2.20 Table

$\times$	1	-1
1	1	-1
-1	-1	1

Is it the only group with two elements? Call the elements e (identity) and a. Filling the table: $e * x = x$ and $x * e = x$ force the first row and column. Now a needs an inverse a' in $\{e, a\}$. It cannot be e, since $e * a = a \neq e$. So $a' = a$, which forces $a * a = e$, and the table is Table 2.21.

2.21 Table

$*$	e	a
e	e	a
a	a	e

Relabelling 1 as e and -1 as a in Table 2.20 gives Table 2.21. So the two groups have matching operations after renaming: they are *isomorphic*. The notation is $\langle \{1, -1\}, \times \rangle \simeq \langle \{e, a\}, * \rangle$.

The same reasoning shows that there is essentially one group with three elements. For a three-element group with identity e and elements $\{e, a, b\}$, the table is forced to be Table 2.25, after completing Table 2.24. The elements of a group with this table are the three matrices

$$e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$a = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

$$b = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

under matrix multiplication; computing the products confirms the table. Any other three-element group, relabelled so that its identity is e and its other elements are a and b, has the same table. So any two groups of three elements are isomorphic.

2.24 Table
(partly filled)

$*$	e	a	b
e	e	a	b
a	a		
b	b		

2.25 Table
(completed)

$*$	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

The book uses the phrase *up to isomorphism*: "there is only one group of three elements, up to isomorphism." The general question, to determine all groups of a given order n up to isomorphism, is hard. Exercise 20 (not in these notes) shows that there are exactly two groups of order 4 up to isomorphism.

2.9 Group isomorphisms

The tables for $\{1, -1\}$ and $\{e, a\}$ match after relabelling. We need a precise definition that does not rely on writing tables.

DEFINITION 2.22 Let $\langle G_1, *_1 \rangle$ and $\langle G_2, *_2 \rangle$ be groups and $f : G_1 \to G_2$. f is a **group isomorphism** if:

1. f is one-to-one and maps onto G_2; and
2. for all $a, b \in G_1$, $f(a *_1 b) = f(a) *_2 f(b)$ (the **homomorphism property**).

Condition 1 relabels the elements of G_1 by elements of G_2. Condition 2 says that the operations match after the relabelling. When there is an isomorphism from G_1 to G_2, G_1 is **isomorphic** with G_2. The inverse of an isomorphism is again an isomorphism (Exercise 44), so isomorphism is symmetric. To show two groups are isomorphic, we may build an isomorphism in either direction.

Example 2.23. $\mathbb{Z}$ and $2\mathbb{Z}$ (the even integers) are isomorphic under addition. Let $f(m) = 2m$.

- One-to-one: $2a = 2b$ implies $a = b$.
- Onto $2\mathbb{Z}$: an even y has $y = 2c$ for some $c \in \mathbb{Z}$, and $y = f(c)$.
- Homomorphism: $f(a + b) = 2(a + b) = 2a + 2b = f(a) + f(b)$.

So f is an isomorphism. The inverse $f^{-1}(x) = x/2$ is also one.

2.10 Properties of group tables

From Table 2.21 we can list necessary conditions that a table on a finite set must meet to give a group. There must be an identity e: the row opposite e repeats the head row in order, and the column under e repeats the head column in order. Since each a has a left and a right inverse, e appears in every row and every column.

More is true. By Theorem 2.17, the equations $a * x = b$ and $y * a = b$ have unique solutions for every b. So:

RESULT (GROUP TABLE TEST)
In the table of a group, each element appears exactly once in each row and exactly once in each column.

Conversely, suppose a table on a finite set has an identity element and every element appears exactly once in each row and column. Then the operation is a group if and only if it is associative. Checking associativity from a table is usually messy. If the operation is given by a characterizing property of $a * b$, associativity is often easy to check, and this is the case most often met.

Applying the test to a three-element set: if the identity is e and the table has the forced entries, the table must be Table 2.25 (up to naming the other two elements). That table is the group in 2.8.

4. Historical notes (background)

Group theory had three roots in the nineteenth century. Algebra of equations: Lagrange studied permutations of the roots of an equation as a tool for solving it. Number theory: Euler studied remainders of powers a^n modulo a prime p, which have group properties. Geometry: attention moved from invariants of geometric transformations to the transformations themselves. Gauss's work on quadratic forms (1800) also gave equivalence classes that behaved like groups under composition. Walther von Dyck and Heinrich Weber (1882) combined these roots and gave the clear definition of an abstract group.

Abelian groups are named after Niels Henrik Abel (1802–1829). Abel showed that the roots of a solvable equation are permuted by commuting functions, so the groups of those permutations are commutative. Camille Jordan (1870) named such groups *abelian*, and the name is now used for all commutative groups. Abel returned to Norway in 1827 with no position and many debts. He died of tuberculosis at 26, two days before August Crelle succeeded in finding him a university position in Berlin.